

Covariate effect (param ~ covariate . shape)

FN 100% FN 50% 0 FP 50% FP 100%

N = 80

|               |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------------|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| CL~BW.power   | 0 | 0 | 1  | 1  | 0  | 1  | 0  | 2  | 47 | 42 | 39 | 44 | 35 | 29 | 41 | 45 |
| CL~BW.lin     | 0 | 0 | 0  | 0  | 0  | 1  | 1  | 1  | 34 | 25 | 26 | 33 | 18 | 15 | 23 | 22 |
| CL~BMI.power  | 1 | 1 | 1  | 0  | 1  | 1  | 0  | 0  | 8  | 11 | 10 | 8  | 7  | 9  | 10 | 12 |
| CL~BMI.lin    | 0 | 0 | 0  | 0  | 2  | 0  | 0  | 1  | 3  | 7  | 3  | 4  | 3  | 3  | 2  | 3  |
| CL~CRCL.power | 2 | 1 | 2  | 1  | 19 | 18 | 16 | 25 | 1  | 0  | 1  | 0  | 24 | 27 | 26 | 27 |
| CL~CRCL.lin   | 0 | 0 | 1  | 0  | 19 | 18 | 15 | 24 | 1  | 0  | 2  | 0  | 22 | 25 | 19 | 22 |
| CL~SEX.cat    | 1 | 2 | 1  | 1  | 1  | 1  | 0  | 1  | 3  | 2  | 1  | 1  | 2  | 3  | 1  | 3  |
| CL~RACE.cat   | 1 | 0 | 1  | 3  | 0  | 0  | 0  | 1  | 4  | 3  | 1  | 1  | 0  | 2  | 4  | 1  |
| VC~BW.power   | 1 | 2 | 30 | 22 | 1  | 2  | 29 | 22 | 1  | 0  | 30 | 41 | 0  | 3  | 32 | 31 |
| VC~BW.lin     | 1 | 0 | 28 | 13 | 0  | 1  | 21 | 12 | 0  | 0  | 22 | 20 | 1  | 2  | 24 | 22 |
| VC~BMI.power  | 1 | 1 | 2  | 7  | 0  | 0  | 6  | 10 | 1  | 1  | 7  | 12 | 3  | 2  | 8  | 5  |
| VC~BMI.lin    | 0 | 0 | 0  | 4  | 1  | 0  | 5  | 3  | 0  | 0  | 1  | 9  | 1  | 1  | 3  | 8  |
| VC~CRCL.power | 0 | 1 | 3  | 0  | 3  | 1  | 1  | 3  | 2  | 1  | 1  | 2  | 2  | 1  | 1  | 0  |
| VC~CRCL.lin   | 1 | 2 | 2  | 0  | 1  | 2  | 1  | 3  | 0  | 2  | 1  | 1  | 1  | 0  | 1  | 0  |
| VC~SEX.cat    | 0 | 3 | 0  | 4  | 1  | 3  | 0  | 4  | 1  | 0  | 2  | 3  | 0  | 3  | 3  | 6  |
| VC~RACE.cat   | 3 | 1 | 2  | 2  | 0  | 2  | 3  | 2  | 2  | 2  | 2  | 1  | 1  | 0  | 2  | 1  |

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Simulation scenario